



Towards Human-Centered RAG: A Study of AI-Supported Testing Practices in Italian Public Administration

Lina Alrabie
Politecnico di Milano
Milan, Italy
lina.alrabie@polimi.it

Salvatore Andolina
Politecnico di Milano
Milan, Italy
salvatore.andolina@polimi.it

Abstract

This study examines the integration of a Retrieval-Augmented Generation (RAG)-based AI testing assistant within an Italian public administration, focusing on its practical benefits and challenges. Qualitative feedback from 15 software testers reveals that while the assistant improves testing efficiency and supports knowledge retrieval, challenges regarding accuracy, consistency, and transparency reduce user trust and effectiveness. Participants highlighted the need for enhanced data curation, improved interaction, and finer-grained control over the system, indicating areas for future improvements to the assistant and its surrounding workflows. The findings emphasize the importance of adopting a human-centered approach in the design and integration of AI solutions in the public sector, offering valuable insights for the ongoing development of human-centered RAG research.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Large Language Models, Retrieval Augmented Generation, Artificial intelligence, Human-centered AI, Software testing, LLM, HCAI

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1 Introduction

Artificial intelligence (AI) is being adopted across a growing range of organisational settings to streamline processes and support knowledge-intensive work, with a strong impact on humans and society [2]. Large language models (LLMs), in particular, have demonstrated the potential to support workers in a variety of tasks by answering natural-language queries [24], synthesising ideas [9], and generating complex content [12]. Recently, organizations have increasingly begun to adopt RAG systems to expose their internal knowledge bases through natural language interfaces. Compared

to standalone LLMs, RAG offers improved reliability and transparency by grounding generated answers in retrieved content from trusted sources and supporting attribution of generated outputs to specific source documents [10, 16]. This makes it particularly relevant for public administrations, where accuracy, consistency, and transparency are critical [18]. As a result, many such organizations are increasingly relying on the enhanced reliability offered by the RAG approach to exploit the potential of LLMs in supporting their operations. By retrieving information from trusted internal sources, RAG helps mitigate typical LLM challenges such as domain mismatches, factual inaccuracies, and hallucinations, without the need for extensive retraining [11].

While the RAG approach offers clear potential for improving information access and task support in public sector contexts, it still presents several challenges, with concerns regarding trustworthiness, system transparency, and the validity of retrieved and generated content [17, 18]. Moreover, effectively integrating such systems into real-world environments requires careful alignment with users' workflows, organizational practices, and domain-specific needs [14, 19]. We argue that addressing these challenges requires turning current RAG systems into Human-centered RAG systems, that is, RAG systems that draw on principles from Human-Centered AI and related movements to ensure better alignment with human values and needs [8, 20]. In this study, we gather initial insights on the path to follow to realize that vision.

1.1 AI in Public Administration

Recent research highlights that effective AI adoption in public administration depends not only on technological capabilities but also on the integration of both technical and non-technical human competencies, which are essential for public value creation [25]. The literature identifies persistent challenges, including tensions around trust, transparency, and ethical issues [18]. A user study with 50 financial professionals by Ravi et al. [21] on a user-centric RAG system for financial equity research found that, while confidence indicators offered useful context, they had limited impact on user trust. In contrast, features that targeted source transparency and user control, such as document filtering and citation highlighting, significantly improved trust and interpretability. These early findings, though not yet definitive, provide a foundation to build upon and offer actionable design guidance for deploying RAG systems in high-stakes, expert-driven domains. Overall, although AI continues to shape workflows, public administration research must confront its associated challenges through interdisciplinary, human-centered approaches [7, 19], emphasizing the importance of designing systems that reflect users' diverse needs, behaviors, and backgrounds [4]. In this regard, the Italian public administration has served as a



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valuable testbed for examining how human-centeredness can be more effectively considered in real-world settings that impact a broad population [7]. In a similar way, we examined AI integration practices in the Italian setting to gather initial evidence on the benefits and practical challenges involved in such integration, with a particular focus on how to design human-centered RAG systems for software testing.

1.2 AI-Supported Software Testing

Recent advances in RAG systems have positioned them as promising tools for enhancing software testing by combining LLMs with structured external knowledge. This integration enables the analysis of extensive codebases and the detection of complex patterns often overlooked by traditional methods, facilitating the generation of more comprehensive test cases [1]. Shin et al. [22] studied the use of RAG systems for unit test generation, using API documentation, GitHub issues, and Stack Overflow content as knowledge sources. They found that the type of source affects the quality of the generated tests. In particular, using GitHub issues can greatly improve code coverage, but system efficiency decreases when the number of test cases is limited. Wang et al. introduced "Copilot for Testing," a context-aware RAG-based testing assistant, which dynamically adapts to real-time code changes. Evaluations demonstrate significant improvements in bug detection accuracy and critical test coverage [27]. In the RAGTAG approach, test scenarios were automatically derived from bilingual requirements in an industrial setting, achieving promising results in relevance and feasibility despite challenges in capturing action-specific details [3].

Despite these advances, AI adoption in testing practice remains limited. A systematic mapping study by Karhu et al. [13] found that real-world applications of AI in testing activities are still sparse, with a significant gap between theoretical promise and practical implementation. The study identified test case generation, intelligent automation, and code analysis as primary use cases, but noted a lack of empirical validation in industrial contexts.

Ultimately, AI-supported software testing remains underexplored and often lacks a deep consideration of human expectations and needs. In contrast, AI applications in public administration are increasingly shaped by expectations of oversight, trust, and transparency. This disconnect between the capabilities of current AI testing tools and the demands of public service environments motivates our investigation of RAG integration practices in the Italian public administration.

2 User study

To investigate how AI-supported testing systems are perceived and used in real-world settings, we conducted a user study involving professional software testers within an Italian public administration context who regularly use an AI-supported testing system in their work. The study aimed to capture both the practical benefits and limitations of a RAG-based assistant, focusing on how such systems align with testers' expectations, workflows, and trust requirements.

2.1 System Overview

The system under study is an AI-powered assistant developed to support software testing activities within public administration

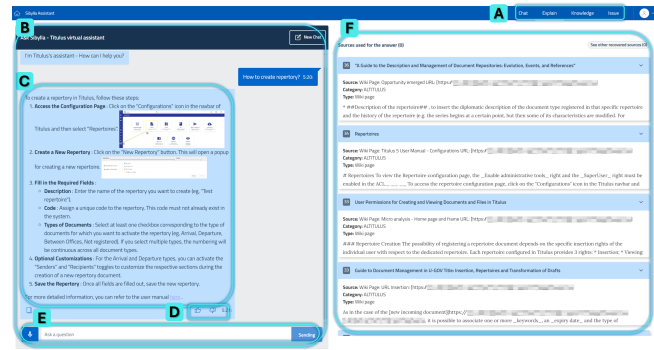


Figure 1: The conversational user interface, featuring: (A) system tabs for managing different users' tasks; (B) a chat panel; (C) a generated response to the user query including embedded visual guidance and a link to relevant documentation for further details; (D) feedback icons for user engagement; (E) the prompting area for text and voice input; and (F) an explainability panel showing the sources used for the generated response.

settings. It is a domain-specific RAG system that allows testers to interact with organizational documentation through a natural language interface. It retrieves information from a curated, local knowledge base, and synthesizes responses using LLMs.

Key features of the system include:

- A conversational interface for querying documentation and test-related information;
- Access to the retrieved sources to support answer verification
- Facilitate the integration and management of domain-specific knowledge sources, supporting various file formats.
- Ensure data ownership and confidentiality for generated responses.

Functionalities and User Interface. This AI assistant was developed to serve different user roles, including testers, knowledge managers, and system administrators. Its user interface provides functionalities that support the tasks associated with each profile (Figure 1).

- (1) **Chat tab:** The main interaction page (Figure 1B), where users can enter queries via text input or use the voice icon to prompt the assistant through speech (Figure 1E). The generated responses may include visual guidance and links to relevant documentation (Figure 1C).
- (2) **Explainability feature:** Allows users to view and assess the sources used to generate responses (Figure 1F). This feature is meant to increase transparency to and foster critical assessment of responses.
- (3) **Knowledge tab:** Available for knowledge managers to explore how information is distributed and clustered within the knowledge domain for a specific topic. This tab also enables adding, modifying, deleting, and searching content.
- (4) **Issue tab:** Integrated with Jira,¹ which is a project management and issue-tracking tool used by teams to plan, track,

¹<https://www.atlassian.com/software/jira>

and manage tasks, activities, or service requests. This feature supports testers in generating summaries and test cases from existing issue tickets. It is available only on the testers' interface.

- (5) **Feedback mechanism:** Allows users to rate the quality and usefulness of the generated responses, supporting iterative improvement of the system (Figure 1D).

2.2 Methodology

This study adopts a case study approach to examine current attempts to introduce RAG systems in public administration settings. Specifically, we investigate how software testers interact with AI-supported testing processes, using an existing tool designed to support testing activities in an Italian public administration. The goal is to surface the challenges and opportunities of applying RAG systems to expose organizational knowledge in a way that aligns with human needs and supports daily workflows.

2.2.1 Participants and Study Design. The study involved 15 system testers, working in the IT department of the administration. The volunteer participants were diverse in terms of age, gender, qualifications, and years of experience (see Table 1).

Data collection focused on qualitative data:

- **Survey Phase:** All participants completed a structured online questionnaire, which addressed the dimensions of system usage, perceived accuracy, trust, transparency, explainability, and areas for improvement.
- **Interview Phase:** To explore more deeply the themes that emerged from the survey phase, four participants volunteered for follow-up, semi-structured interviews. Each interview lasted approximately 30 minutes and was conducted via a video conferencing platform.

2.2.2 Data Analysis. We employed a qualitative data analysis strategy. Survey responses and interview transcripts were subjected to *thematic analysis*, following Braun and Clarke's six-step data analysis procedure [6].

A combination of **MAXQDA** and **Microsoft Excel** was used to support iterative coding, theme development, and cross-source triangulation. Codes were developed inductively and then clustered into 13 higher-level themes. These themes captured both perceived advantages and recurring challenges related to the use of the AI system under study, as well as user-suggested areas for improvement.

The final coding schema included the following categories:

- **Advantages:** Speed and Efficiency (ADV01), Knowledge Support (ADV02), Generative Support (ADV03).
- **Challenges:** Hallucinations (CHAL01), Specificity Limitations (CHAL02), Trust and Verification (CHAL03), Documentation Quality (CHAL04), Consistency Issues (CHAL05), Explainability (CHAL06), Source Transparency (CHAL07).
- **Improvement Suggestions:** Improve Data Ingestion (IMP01), Better UI/UX (IMP02), Improve Documentation (IMP03).

This structure enabled the synthesis of general trends and participant insights into a cohesive understanding of how testers engage with RAG-based systems, highlighting both the system-level limitations and human-centered design opportunities.

3 Findings

This section presents the key findings derived from our study on the perceptions and experiences of public administration personnel regarding AI-supported testing practices. The findings are categorized into perceived advantages, identified challenges, and suggested improvements, each supported by direct quotes from participants.

3.1 Perceived Advantages of the AI-Supported Testing System

Participants consistently highlighted several benefits of integrating AI into their testing workflows, primarily revolving around enhanced efficiency and knowledge acquisition.

3.1.1 Speed and Efficiency. The AI-supported testing system was perceived as a significant accelerator in routine tasks, enabling users to retrieve and process information with increased speed. This efficiency directly contributed to time savings when navigating documentation and retrieving relevant content.

"Faster times to find information [P03]."

"It can be helpful in decreasing the time spent searching for information in the documentation [P04]."

3.1.2 Knowledge Support. The system proved valuable in bridging knowledge gaps, particularly when users encountered unfamiliar domains, products, or software functionalities. It served as a tool for initial understanding and for outlining complex processes. Participants reported gaining "information on products that were not known [P08]," and recognizing its utility as "an integration tool to outline the steps of a test; integrative knowledge with respect to software areas of which one does not have mastery; introduction to the main functions of a software with concise and clear explanations [P07]."

"You can get some starting information on topics that are not known [P09]."

3.1.3 Generative Support. A key advantage identified was the system's capacity to generate or guide the creation of test plans, test scenarios, and queries. This generative capability streamlined the initial phases of test development. For instance, participants appreciated:

"The instant creation of a test plan, support for creating test cases [P01]."

"The most interesting is the construction of database queries [P06]."

3.2 Identified Challenges in AI-Supported Testing

Despite these advantages, several critical challenges were reported, primarily related to the reliability and contextual precision of the AI-generated outputs.

3.2.1 Hallucinations. A prevalent issue was the occurrence of "hallucinations," in which the system provided factually incorrect or misleading information. This significantly undermined confidence in the system's reliability. Participants frequently encountered "too frequent hallucinations [P05]," with some responses being "completely wrong [P01]."

Table 1: Study participants’ demographic and methodological data

ID	Age Group	Gender	Education	Experience	Study Tool
P01	41+	Male	Bachelor’s degree	11–20 years	Questionnaire
P02	41+	Female	High school, diploma or equivalent	6–10 years	Questionnaire
P03	31–40	Female	Master’s degree	2–5 years	Questionnaire & Interview
P04	31–40	Male	Master’s degree	2–5 years	Questionnaire
P05	41+	Male	Master’s degree	11–20 years	Questionnaire & Interview
P06	41+	Female	High school, diploma or equivalent	More than 20 years	Questionnaire & Interview
P07	31–40	Female	Bachelor’s degree	2–5 years	Questionnaire
P08	31–40	Female	Master’s degree	6–10 years	Questionnaire
P09	41+	Male	Bachelor’s degree	2–5 years	Questionnaire
P10	41+	Female	Master’s degree	More than 20 years	Questionnaire & Interview
P11	31–40	Male	High school, diploma or equivalent	6–10 years	Questionnaire
P12	20–30	Male	High school, diploma or equivalent	2–5 years	Questionnaire
P13	41+	Female	Bachelor’s degree	More than 20 years	Questionnaire
P14	41+	Female	Master’s degree	More than 20 years	Questionnaire
P15	41+	Male	Master’s degree	More than 20 years	Questionnaire

"It often happens that among the answers there are hypotheses on functions of the system, which in most cases turn out to be wrong [P04]."

3.2.2 Specificity Limitations. The system’s responses were often found to be overly generic, lacking the detailed actionable guidance required for specific testing activities or resolving operational impasses. Users frequently received "answers too generic [P15]," which limited the practical applicability of the information.

"Often you get more of a general smattering, and it does not always allow you to overcome any operational impasses in the testing activity or in accessing the environments [P08]."

"Sometimes, the steps suggested to reproduce the test case were generic, skipping some steps necessary to correctly replicate the test [P11]."

3.2.3 Trust and Verification. A fundamental challenge was the inherent lack of immediate trust in the AI’s output, necessitating extensive user validation. Participants consistently felt the need to verify the resulting information, often by consulting external sources such as ChatGPT, Claude, and Gemini, or relying on their own expertise and colleagues. This need for continuous oversight imposed a cognitive burden on users, as reflected in statements like:

"You need to verify if the answers provided are correct [P04]."

"The information received still needs to be verified [P02]."

3.2.4 Documentation Quality. The quality of the system’s responses was observed to be heavily dependent on the quality of the underlying documentation it accessed. Issues in source materials directly led to inaccurate or inconsistent outputs.

"One main difficulty of the system is dependence on existing written documentation [P08]."

"Answers that are not consistent with the question or incorrect due to deficient information sources [P07]."

3.2.5 Consistency Issues. Participants reported instances where the system provided different answers to the same or very similar questions, leading to confusion and reduced reliability. This manifested as "inconsistent information on the same topic [P09]."

"Information received is not always precise and in relation to what was requested [P02]."

"To the same question, I get different answers [P10]."

3.2.6 Explainability. Despite the system’s support for explanations, transparency remained a significant concern. Approximately 80% of participants indicated that the system’s explanations were unclear or in need of improvement, which affected their trust in the system.

"Explanations have to be improved [P10]."

"I don’t understand if the system has selected the correct information and based on which documents [P05]."

3.2.7 Source Transparency. A key concern expressed by participants regarded the need to better identify the specific sources used by the RAG system to generate responses. Instead of referencing precise documents or sections, the system often provided only vague or generic links. As participants noted:

"It should indicate the specific information sources used to elaborate the answer and not just, for example, the parent link of the wiki [P07]."

"The pages used for the answer should be specified [P08]."

3.3 Suggested Improvements for AI-Supported Testing

Based on their experience, users identified several areas for improving the AI-supported testing system, with particular emphasis on data ingestion processes, user interface design, and the quality of the underlying documentation.

3.3.1 Improve Data Ingestion. A key aspect for improvement was the process by which the AI system selects and ingests data from its sources. Suggestions included the continuous enhancement of source quality and relevance.

"Improve the data selection and ingestion process [P15]."

"Find a way to improve the answers... and select quality sources [P01]."

3.3.2 Better UI/UX. For a better user experience, feedback highlighted the importance of improving the intuitiveness of the user interface, facilitating interactions by minimizing complexity, especially in cases where applications are accompanied by extensive documentation.

"Make the interface a little more intuitive [P03]."

"The user interface can be complex in some applications with vast documentation [P06]."

3.3.3 Improve Documentation. A recurring suggestion highlighted the critical need to address issues with the underlying documentation itself, including outdated content, information overload, or lack thereof. This was seen as fundamental to improving the system's output quality. One participant stressed that it is a promising improvement to "use clear, schematic, and comprehensive information sources" and noted that "reviewing and correcting the documentation is an expensive ad hoc job [P07]."

"Eliminate background information, and select only sources within the documentation that are consistent with the request (e.g., if there are pages with the same name or similar content, the system gets confusing) [P08]."

"It needs continuous documentation updates to provide more precise answers [P11]."

4 Discussion

This study provides empirical insights into the practical application of RAG-based AI systems within a public administration testing context, revealing both benefits and systemic limitations. Testers recognized the system's value as a time-saving tool that improves access to information and supports task fulfillment. These advantages align with broader expectations of RAG systems in organizational environments, particularly their capability to accelerate knowledge retrieval and lower cognitive entry barriers for complex tasks.

However, the findings also reveal several key challenges that impact the effectiveness and trustworthiness of the analyzed AI-supported workflow. Persistent issues that are well-documented in the literature, such as hallucinations, overly generic outputs, and inconsistent responses, highlight the ongoing tension between generative fluency and domain-specific accuracy in RAG systems [18, 21]. Users often had to compensate for these shortcomings through manual verification, drawing on their own domain knowledge, external tools, or input from colleagues. This not only lowered expected efficiency gains but also added to users' cognitive workload, as they had to remain actively engaged to detect and correct system errors due to its perceived unreliability [15].

High-quality documentation emerged as a critical factor for both the accuracy of AI-generated output and for sustaining user trust. When the documentation was incomplete or outdated, the model tried to fill the gaps, often inventing navigation steps that didn't exist. This could be particularly problematic during onboarding, where subtle hallucinations are easy to miss, compromising the very benefits of the RAG-based testing assistant. Our findings suggest that such issues decrease trust, and as a consequence, testers typically revert to standard onboarding procedures, seeking help from more-senior colleagues instead of relying on the assistant.

Moreover, transparency and control emerged as key areas for improvement. Participants stressed the need for clear source references and insight into the system's internal processes to assess answer credibility and build trust. They also sought greater control over source selection, which would enhance the sense of agency when using the system [21].

To address these challenges, participants provided several useful suggestions, including enhancing data ingestion processes to cut noise and improve relevance, updating documentation regularly, and redesigning the user interface to enhance user experience.

Taken together, these findings show that successful AI integration in public administration may not only depend on technical accuracy but also on the optimization of internal processes [5, 26] as well as on better consideration of human-centeredness in the design of AI-infused systems [2, 4, 8]. They highlight the need to reconsider current AI system designs through the lens of Human-Centered AI principles [20, 21, 23], and propose a path toward designing meaningful interactions between humans and AI, advancing the vision of achieving truly human-centered RAG systems.

5 Conclusions

In this paper, we employed a human-centered approach to understand the challenges of integrating a RAG-based AI testing assistant in an Italian public administration. The study aimed to identify the assistant's practical value, limitations, and improvement opportunities. We collected qualitative data from 15 participants and found that while the system enhances efficiency and supports knowledge discovery, its limitations in accuracy, specificity, and explainability significantly undermine user trust and effectiveness. The challenges identified, particularly those around hallucinations, documentation dependence, and transparency, reflect broader issues in the adoption of generative AI in complex domains such as public administration. Ultimately, this study highlights the importance of embedding human-centered design principles in all AI integration efforts. Doing so will not only improve usability and trust, but also ensure better human-AI interaction and stronger compliance with ethical guidelines in high-stakes institutional settings.

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